

Supplementary Material: Lite Learning: Efficient Crop Classification in Tanzania Using Feature Extraction with Machine Learning & Crowd Sourcing

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S-I. TIME SERIES FEATURE DEFINITIONS

This document is the online supplement to the main article “*Lite Learning: Efficient Crop Classification in Tanzania Using Feature Extraction with Machine Learning & Crowd Sourcing.*” It provides the full list of time-series features referenced in the main text (Methods, “Time Series Features”).

The features were extracted on a per-pixel basis from the 2023 Sentinel-2 monthly composites (bands B2, B6, B11, B12, the Enhanced Vegetation Index (EVI), and hue) using the `xr_fresh` toolkit [1]. They summarize the temporal dynamics of crop growth and development across the growing season, spanning basic statistical descriptors, changes over time, and distribution-based metrics. Table S1 lists each statistic, a short description, and its defining equation. Throughout, x_i denotes the value of the band or index at time step i in a pixel’s monthly time series of length n , $\bar{x}$ its mean, σ its standard deviation, μ_4 its fourth central moment, and l an autocorrelation lag.

TABLE S1: Time-series features extracted with `xr_fresh`, with descriptions and defining equations.

Statistic	Description	Equation
Absolute energy	sum over the squared values	$E = \sum_{i=1}^n x_i^2$
Absolute Sum of Changes	sum over the absolute value of consecutive changes in the series	$\sum_{i=1}^{n-1} x_{i+1} - x_i $
Autocorrelation (1, 2 & 3 month lag)	Correlation between the time series and its lagged values	$\frac{1}{(n-l)\sigma^2} \sum_{t=1}^{n-l} (x_t - \bar{x})(x_{t+l} - \bar{x})$
Count Above Mean	Number of values above the mean	$N_{\text{above}} = \sum_{i=1}^n (x_i > \bar{x})$
Count Below Mean	Number of values below the mean	$N_{\text{below}} = \sum_{i=1}^n (x_i < \bar{x})$
Day of Year of Maximum Value	Day of the year when the maximum value occurs in series	—
Day of Year of Minimum Value	Day of the year when the minimum value occurs in series	—
Kurtosis	Measure of the tailedness of the time series distribution	$G_2 = \frac{\mu_4}{\sigma^4} - 3$

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Statistic	Description	Equation
Linear Time Trend	Linear trend coefficient estimated over the entire time series	$b = \frac{\sum_{i=1}^n (t_i - \bar{t})(x_i - \bar{x})}{\sum_{i=1}^n (t_i - \bar{t})^2}$
Longest Strike Above Mean	Longest consecutive sequence of values above the mean	—
Longest Strike Below Mean	Longest consecutive sequence of values below the mean	—
Maximum	Maximum value of the time series	$x_{\max}$
Mean	Mean value of the time series	$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$
Mean Absolute Change	Mean of absolute differences between consecutive values	$\frac{1}{n-1} \sum_{i=1}^{n-1} x_{i+1} - x_i $
Mean Change	Mean of the differences between consecutive values	$\frac{1}{n-1} \sum_{i=1}^{n-1} (x_{i+1} - x_i)$
Mean Second Derivative Central	measure of acceleration of changes in a time series data	$\frac{1}{2(n-2)} \sum_{i=1}^{n-2} \frac{1}{2} (x_{i+2} - 2 \cdot x_{i+1} + x_i)$
Median	Median value of the time series	$\tilde{x}$
Minimum	Minimum value of the time series	$x_{\min}$
Quantile (q = 0.05, 0.95)	Values representing the specified quantiles (5th and 95th percentiles)	$Q_{0.05}, Q_{0.95}$
Ratio Beyond r Sigma (r=1,2,3)	Proportion of values beyond r standard deviations from the mean	$P_r = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x} > r\sigma)$
Skewness	Measure of the asymmetry of the time series distribution	$\frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{\sigma} \right)^3$
Standard Deviation	Standard deviation of the time series	$\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$
Sum Values	Sum of all values in the time series	$S = \sum_{i=1}^n x_i$
Symmetry Looking	Measures the similarity of the time series when flipped horizontally	$ \bar{x} - \tilde{x} < r \cdot (x_{\max} - x_{\min})$
Time Series Complexity (CID CE)	measure of number of peaks and valleys	$\sqrt{\sum_{i=1}^{n-1} (x_{i+1} - x_i)^2}$
Variance	Variance of the time series	$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$
Variance Larger than Standard Deviation	check if variance is larger than standard deviation	$\sigma^2 > 1$

S-II. COMPUTATIONAL COMPLEXITY OF DEEP-LEARNING BASELINES

This section expands the efficiency comparison in the main text (Methods, “Computational Efficiency”), where the LightGBM “lite learning” model trains in 10–15 minutes on a CPU-only laptop. By contrast, the deep-learning architectures commonly applied to satellite image time series carry substantially higher training-time and memory costs. As in the main text, n denotes the number of training samples.

Recurrent models such as LSTMs, frequently used for temporal land cover analysis, exhibit complexity:

$$T_{LSTM} = O(e \cdot n \cdot t \cdot (h^2 + h \cdot i))$$

where e is the number of epochs, t the sequence length, h the hidden dimensionality, and i the input feature dimension. The quadratic dependence on h and the inherently sequential nature of LSTMs prevent efficient parallelization across the temporal axis, often extending training to several hours.

Spatial architectures further escalate these demands. A 2D U-Net scales linearly with the number of input channels (treating time steps as channels), but a 3D U-Net, which treats the temporal dimension as a third spatial axis, follows:

$$T_{3DUNet} = O\left(e \cdot n \cdot \sum_l k_l^3 \cdot c_{l-1} \cdot c_l \cdot s_l\right)$$

where the sum runs over network layers l with kernel size k_l , input and output channel counts c_{l-1} and c_l , and spatial size s_l . The cubic kernel term (k_l^3) means a $3 \times 3 \times 3$ convolution requires 27 operations per voxel, compared to 9 for a 2D equivalent. Furthermore, the memory (VRAM) required to store 4D tensors (Space $\times$ Time $\times$ Channels) for large-scale mapping routinely necessitates high-end GPU clusters.

REFERENCES

- [1] M. L. Mann, “xr_fresh: Python package for feature extraction from raster data time series,” Zenodo, 2024. [Online]. Available: <https://doi.org/10.5281/zenodo.12701466>